CEFOTAXIME

Preferred in neonates

Cross-allergy

CHEMICAL NAME (6R,7R)-3-(acetyloxymethyl)-7-[[(2Z)-2-(2-amino-1,3-thiazol-4-yl)-2-methoxyiminoacetyl]amino]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (sodium)

CLASS Third-generation cephalosporin (parenteral)

TRADE NAMES Claforan · Taxim · Cefotaxime sodium injection

THERAPEUTIC USES / INDICATIONS

- Neonatal sepsis and meningitis (preferred over ceftriaxone in neonates).
- Bacterial meningitis; septicaemia.
- Lower respiratory tract, urinary and intra-abdominal infections.
- Gonorrhoea; surgical prophylaxis.

ADMINISTRATION

- **IV** injection over 3–5 min or infusion over 20–60 min; **IM**.
- Rapid bolus (<1 min) via central line can cause arrhythmia.
- Compatible with calcium-containing fluids (unlike ceftriaxone).

DOSAGES

- **Adult:** 1 g every 12 h (mild) to 2 g every 6–8 h (severe); max 12 g/day.
- **Meningitis:** 2 g every 4–6 h.
- **Neonates:** 50 mg/kg every 8–12 h (age-dependent).
- **Children:** 50 mg/kg every 8 h (meningitis every 6 h).
- Halve dose if CrCl <5 mL/min.

PHARMACOKINETICS

- **Onset:** immediate IV. Good CSF penetration.
- Metabolised in liver to desacetylcefotaxime (active).
- Excreted in urine. **t½:** 1 h.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Spectrum similar to ceftriaxone: Gram-negatives, pneumococci, meningococci; not *Pseudomonas*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site inflammation, pain.
- Diarrhoea, nausea.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- *C. difficile* colitis.
- Arrhythmia with rapid central injection.
- Neutropenia, eosinophilia (prolonged).

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity; severe penicillin allergy.
- **Caution:** renal impairment.

KEY INTERACTIONS

- **Aminoglycosides:** ↑ nephrotoxicity; give separately.
- **Probenecid:** ↑ levels.

NURSING MANAGEMENT

- Check allergy history.
- Neonates: weight-based dose by postnatal age.
- Give IV over at least 3–5 min.
- Monitor temperature, cultures, CBC, renal function.
- Monitor IV site.

PATIENT EDUCATION

- Report rash or difficulty breathing.
- Report persistent diarrhoea.

KEY POINTS

- 3rd-gen cephalosporin; good CSF penetration.
- Can be given with calcium fluids.
- Preferred over ceftriaxone in neonates.
- No Pseudomonas cover.

SOURCES CHECKED **[138]** Cefotaxime 1 g Powder for solution for injection vial (UK SmPC) **[139]** Cefotaxime, IM, IV (US label) **[140]** Cefotaxime, IV (US label)

Full references p. 432–457